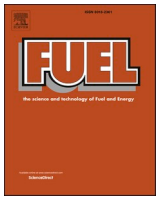
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Full Length Article



Deep eutectic solvents promote in-situ hydrogenolysis of *Evodia lepta* residue in preparing small aromatic phenol ketones

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ABSTRACT

A deep eutectic solvent (DES) system of ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ was developed for simultaneous dissolution of lignin from *Evodia lepta* residue and *in-situ* hydrogenolysis of lignin to aromatic phenol ketones using a Ru-C catalyst. The use of this DES led to a delignification of reached 93.1 % and cellulose and hemicellulose retentions of 95.3 % and 47.8 % respectively at the molar ratio of 1:12:0.2. While the total aromatic phenol ketones yield reached 92.4 % under 130 °C for 3 h with 17.2 % yields of vanillyl methyl ketone, 9.1 % syringylacetone, 25.4 % dihydroeugenol, and 40.7 % 4-propyl-2,6-dimethoxyphenol. The improved Ru-C properties in the DES system like thermal stability, the specific surface area and reduction potential contributed on the exceptional high product yields. The aromatic phenol ketones were 312.5 % higher than the reported yields of hydrogenolysis of aromatic monomers by ethanol/1,4-dioxane/formic acid under 300 °C–2 h conditions.

1. Introduction

Over the past two decades, most traditional biorefineries have focused on converting cellulose and hemicellulose into biofuels, such as ethanol, diesel, and jet fuel [1–3]. However, lignin is used primarily as a low-value fuel due to its complex structure, irregular chemical bonds, and difficult degradation [4]. In recent years, increasing attention has been paid to "lignin-first" strategies, which release lignin from natural biomass, preventing its condensation by catalytic or protective group chemistry before direct conversion is enabled [5]. For instance, Jia et al. [6] used a lignin-first strategy to achieve a 40.4 % monophenol yield from natural lignin under mild conditions (200 °C for 4 h) in the presence of Ru-C and H_2WO_4. Elsewhere, Ma et al. [7] fractionated corn stover into stabilized lignin oil. The lignin was decomposed into oligomers (25.1 % yield), such as p-coumaric and ferulic acid methyl esters, using an H_2WO_4 catalyst at 200 °C for 6 h. Achieving the high-value resource utilization of dregs from biomass. It has a vast potential for developing carbon neutrality [8,9].

Catalytic hydrogenolysis is an attractive approach for depolymerization of lignin into higher-value aromatic compounds [10,11]. For example, Wang et al. [12] used a Ru catalyst for the selective hydrogenolysis of catechin-lignin to aryl catechols (77 %). The authors inferred from model compounds that caffeol produced by C-O bond cleavage may act as an intermediate in the C-lignin hydrogenolysis reaction. Likewise, Kong et al. [13] hydrolyzed industrial lignin at 300 °C for 4 h. Using ethanol as the hydrogen donor solvent, a 20.6 % yield of aromatic monomers was achieved. The catalytic hydrogenolysis of lignin into high-value chemicals and fuels has recorded many breakthroughs. Still, the harsh reaction conditions remain problematic, requiring an exigent approach to construct an efficient solvent-catalyst coupling system [14].

Solvents are crucial in depolymerization, they penetrate the biomass, increasing lignin solubility, inhibiting the formation of undesirable intermediates, and reducing biochar production [15]. Alcohols (such as methanol [16] and ethanol [17]) and non-polar solvents (such as dioxane and cetane [18]) are often used as solvents for lignin depolymerization. In addition, Deep eutectic solvents (DESs) exhibit easy synthesis and low cost, making them the most promising solvent for lignocellulosic biomass processing, especially for lignin value-added solvents [19–21]. Furthermore, DES was more green than traditional solvent. Taking choline chloride/MeOH as an example, ChCl could alter the solubility and volatility of MeOH by solvent effects and molecular

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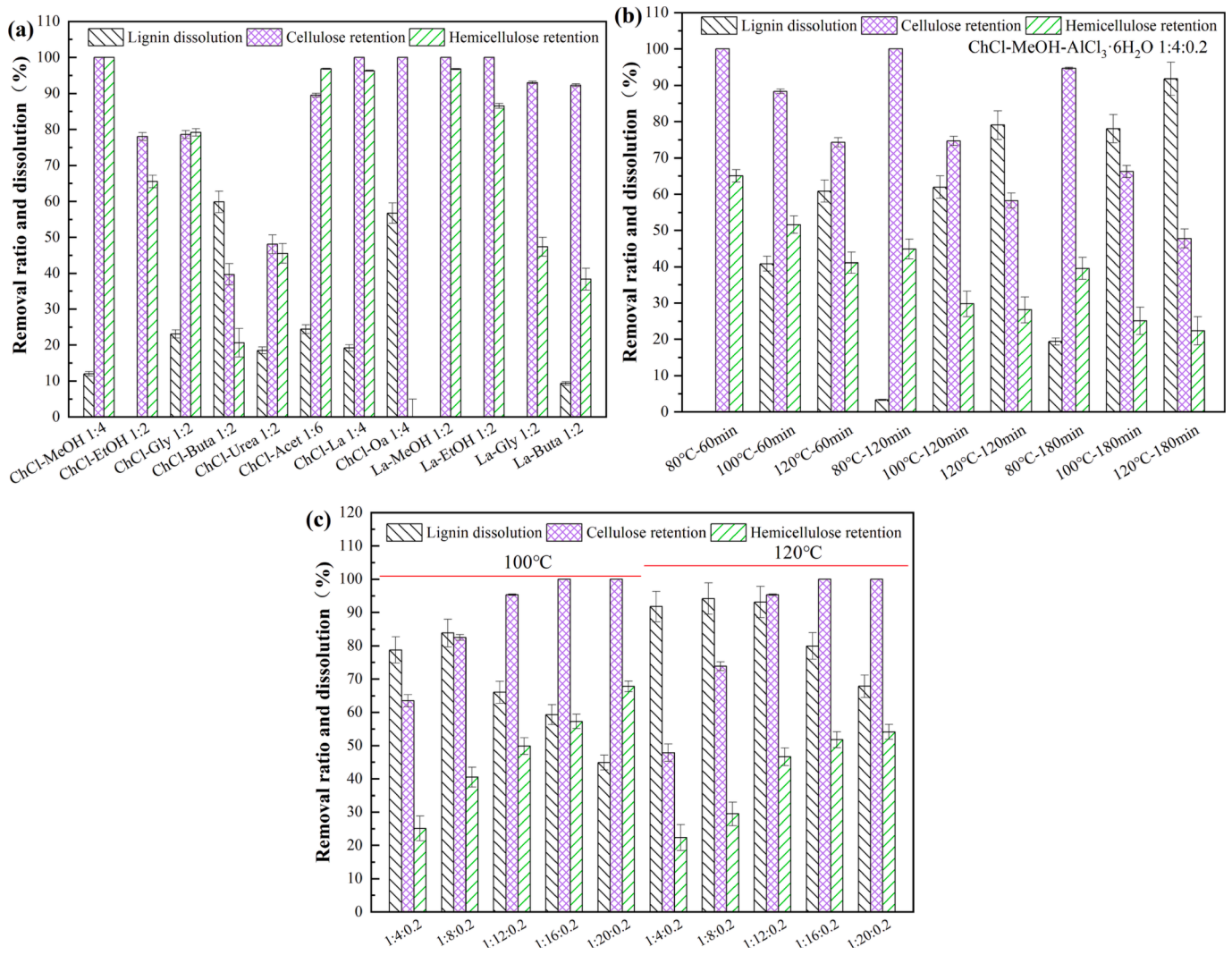


Fig. 1. Effect of DES types (a), reaction temperature–time (b), and molar ratio of ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ (c) on changes in the chemical composition of *Evodia lepta* residue.

forces to make MeOH less toxic. Yu et al. [22] reported that ChCl/glycerol (DES) depolymerized lignin at 170 °C for 50 min, releasing 17 acids with a relative abundance of 80.7 %. Likewise, Wang et al. [23] reported a catalytic depolymerization reaction of ChCl/lactic acid (DES) C-lignin with Pd/C to produce monomeric catechol derivatives in 29.6 wt% yield. These studies illustrate DES's potential to solubilize depolymerized lignin. However, studies on the interaction between DES and the catalysts in the reaction system are lacking. We have previously reported the improved lignin solubility and the redox potential of Cu $(OAc)_2$/1,10-o-phenanthroline catalyst using a ChCl-MeOH-based DES system [24], which led to a total yield of 87.1 % for acetyl vanillin (AV) and acetic acid. In another study, Jian et al. [25] reacted the lignin derivative AV with 4-hydroxybenzaldehyde as substrate. Under ChCl/formic acid (DES), 15.0 wt% H-β, and 100 °C, the DES converted AV from 39.6 % to 88.7 % by increasing the strong acid strength of H-β and increasing the surface area. These studies further illustrate the potential of DES-enhanced catalysts, but little research has been conducted on how DES synergizes with catalysts to regulate the distribution of hydrogenolysis products.

In the current study, various DES types were firstly screened to investigate the selective dissolution of lignin from *Evodia lepta* residue – a by-product from Chinese herbal medicine production. Furthermore, the DES with the highest delignification was coupled with a Ru-C catalyst develop a catalytic system for simultaneous delignification of biomass and *in situ* hydrogenolysis of lignin to produce aromatic phenol ketones. The system researched to improve the conversion efficiency and the yield of aromatic phenol ketones. Various analytical approaches, including SEM, XRD, TG, BET and cyclic voltammetry experiments (CV) was used to explore the properties of catalyst and understand the interaction mechanism between the DES and the catalyst. Relevant theoretical and technical supports have been provided for delignification from *Evodia lepta* residue and its *in situ* hydrogenolysis to phenolic ketone under a deep eutectic solvent environment.

2. Materials and methods

2.1. Materials

The *Evodia lepta* residue from Han Chong Tang, a Chinese pharmacy, was the boiled biomass. Then, it was dried at >50 °C for 24 h. The residue was crushed and sieved to obtain 0.25–0.85 mm particles, used as raw materials for lignin extraction and *in situ* hydrogenolysis. The compositional analysis of *Evodia lepta* residue after fractionation and hydrogenolysis was done using the Laboratory for Renewable Energy Laboratory standard method (LAP, NREL) [26]. The lignin, hemicellulose, and cellulose contents of *Evodia lepta* residue before depolymerization were 15.6 %, 15.6 %, and 36.0 %, respectively.

Choline chloride, levulinic acid, methanol, ethylene glycol, glycerol,

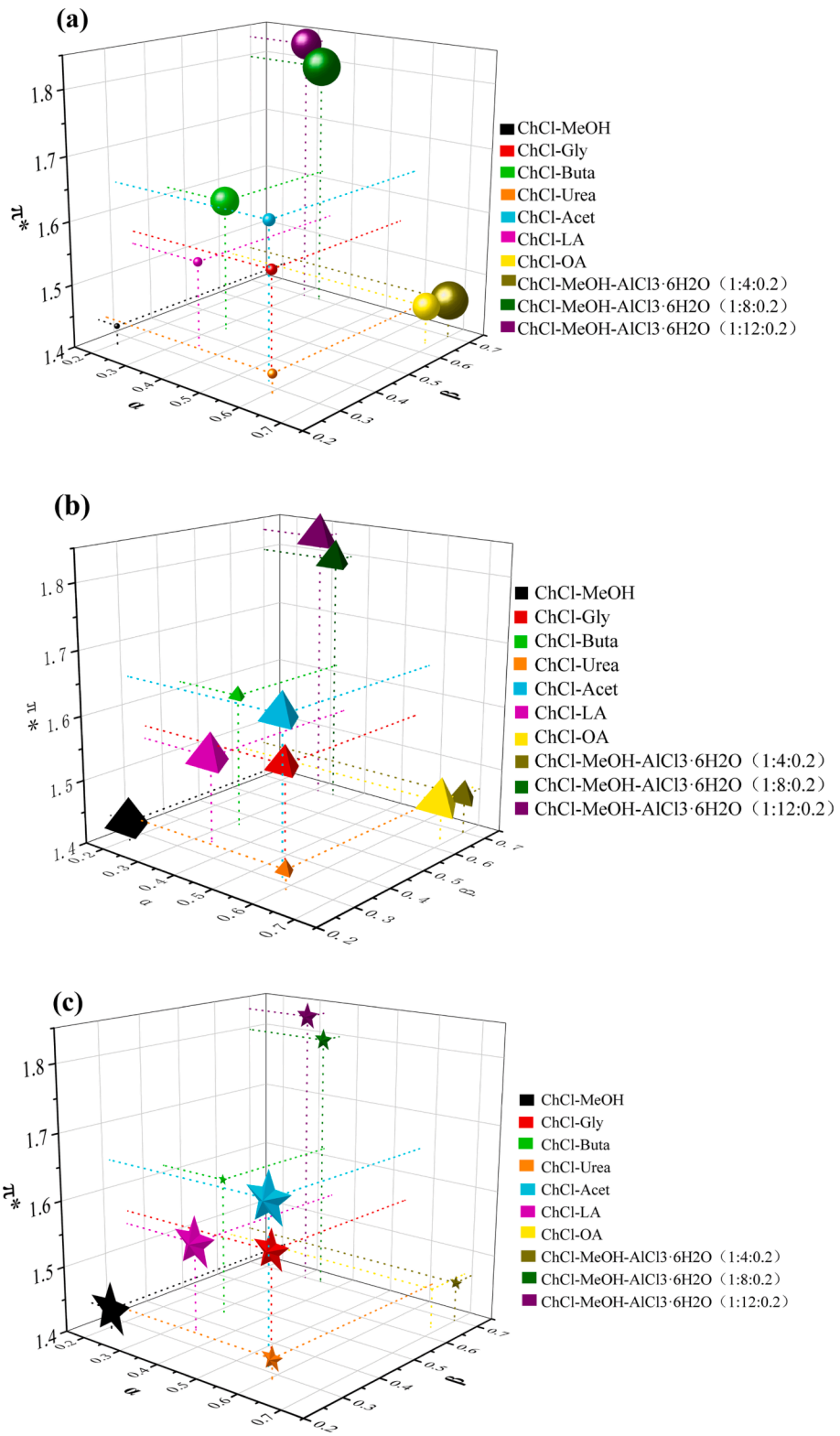


Fig. 2. Correlation analysis of relevant parameters of Kamlet Taft and fractionation of three lignocellulose components: (a) lignin, (b) cellulose, (c) hemicellulose, the size of the figure represents the extent of lignin dissolution and the retention of hemicellulose and cellulose.

1,4-butanediol, acetamide, oxalic acid, and aluminum chloride hexahydrate were purchased from McLean Biochemistry Technology Co. Ltd. (Shanghai, China). Aladdin's Reagent (Shanghai) Ltd supplied industrial catalysts Ru-C, vanillyl methyl ketone, syringylacetone, and dihydroeugenol.

2.2. DES preparation

The DESs were prepared by stirring HBA and HBD at 80 °C for 30–40 min. choline chloride (ChCl) and levulinic acid (La) were used for HBAs, while methanol (MeOH), ethanol (EtOH), glycerol (Gly), 1,4-butanediol (Buta), acetamide (Acet), oxalic acid (Oa), urea, and aluminum chloride

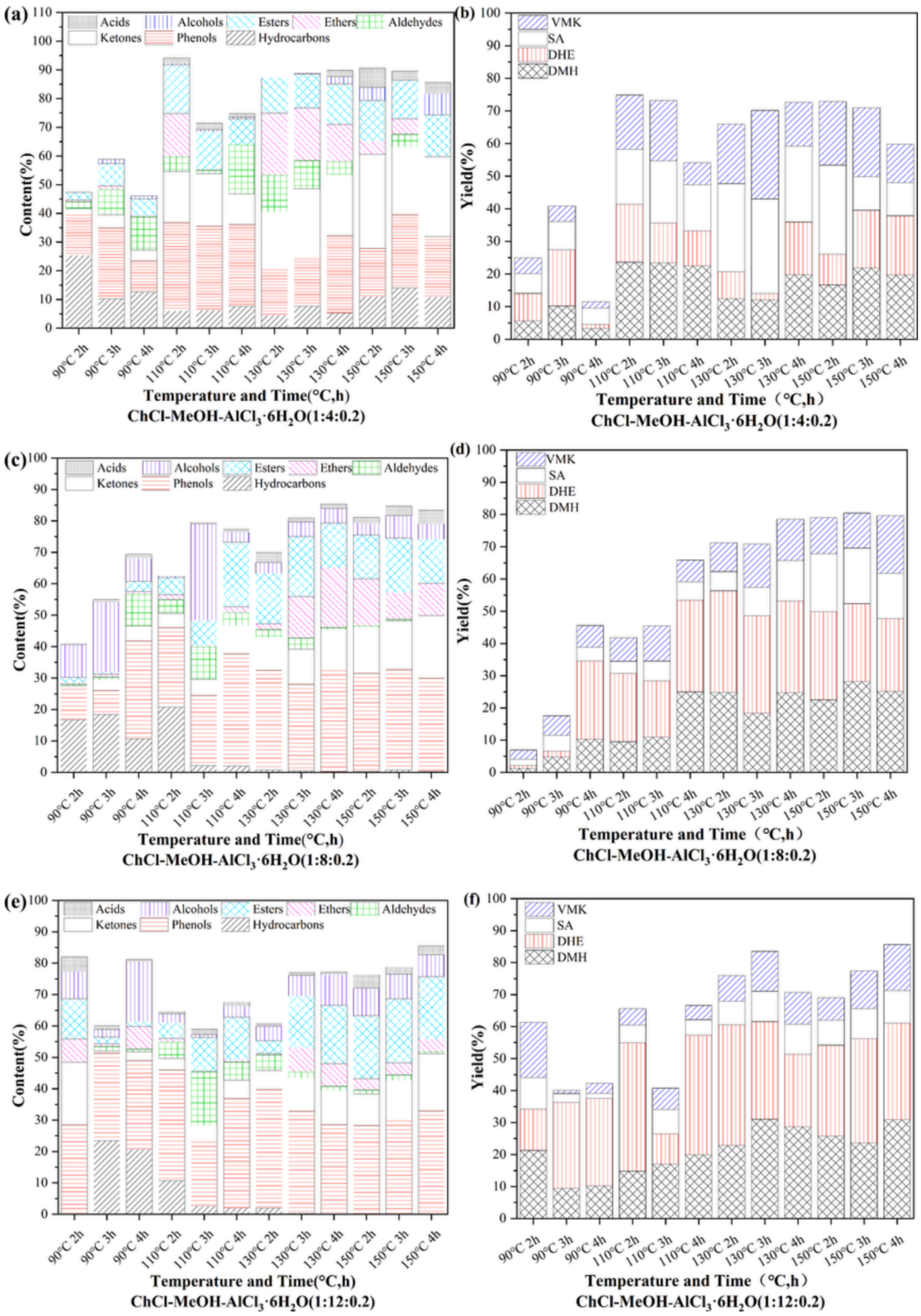


Fig. 3. Effect of temperature and time on the distribution (a) (c) (e) and yield (b) (d) (f) of depolymerization products in ChCl-MeOH-$AlCl_3$·$6H_2O$ system with various molar ratios.

hexahydrate ($AlCl_3{\cdot}6H_2O$) were used for HBDs. Detailed information and the Kamlet-Taft parameters on the thirteen DESs prepared were given in Supporting Information Table.S1 and S1, respectively.

2.3. Selective extraction of lignin

Evodia lepta residue and DES were mixed in a 100 mL reactor (solid–liquid ratio 1:20 g/mL), heated at 100 °C and kept under stirring at 300 rpm for 180 min. After cooling, the liquid was pumped and filtered with a G3 sand-core funnel. An appropriate amount of deionized water was added to the filtrate before being centrifuged for 30 min, and dried to obtain the lignin pretreated with DES. The retained solid on the funnel was dried in an oven at 105 °C for 24 h, before the specific components were determined through compositional analysis.

2.4. Depolymerization of Evodia lepta residue

Evodia lepta residue and DES were mixed in a 100 mL reactor as follows: solid/liquid ratio of 1:20 g/mL; *Evodia lepta* residue/Ru-C mass ratio of 6:1; hydrogen pressure of 3 Mpa; heating at 120 °C; hold-time of 180 min under constant stirring. Details of the tests, such as gas chromatography mass spectrometer (GC–MS), gas chromatograph (GC), high pressure liquid chromatography (HPLC), etc., are provided in Supporting Information S2.

2.5. Characterization of catalysts

A SU-70 instrument (10 kV) obtained the SEM images of the samples. The XRD characterization was performed using an X-ray diffractometer (Smartlab) with the following instrumental parameters: a Cu Kα target radiation, 35 kV tube voltage, 40 mA tube current, and a 5°–80° scanning range. Thermogravimetric analysis (TG) of catalysts was carried out using a fast ramp-up thermogravimetric system (SDT650) under a 40 mL/min N_2 gas stream and a 10 °C/min heating rate. Moreover, a Quantachrome Autosorb IQ3 measured the BET properties of the catalyst. Samples were kept under vacuum at 300 °C for 8 h before testing. The redox potentials of the catalysts with the various DESs were measured using CV experiments. The electrodes used were referenced [24].

3. Results and discussion

3.1. Fractionation of Evodia lepta residue by various DESs

The *Evodia lepta* residue was treated (1:20 solid–liquid ratio, at 100 °C for 180 min) to investigate the lignin solubilization and cellulose and hemicellulose retention by the various DESs. Fig. 1a compares DES with levulinic acid as the hydrogen-bonding acceptor. The DES with ChCl as the hydrogen-bonding acceptor solubilized lignin more effectively, while ChCl-oxalic acid and ChCl-1,4-butanediol DES showed the highest lignin dissolution (56.7 % and 59.8 %, respectively.). Also, the pair displayed low hemicellulose retention, with ChCl-oxalic acid and ChCl-1,4-butanediol DES retaining 0 % and 20.6 %, respectively, compared to other ChCl-based DES, such as ChCl-MeOH with hemicellulose retention of 100 %. ChCl-1,4-butanediol DES was able to retain up to 39.9 % cellulose. Therefore, neither ChCl-oxalic acid nor ChCl-1,4-butanedio DESs could preferentially dissolve lignin from the feedstock. In contrast, only the lignin fraction was removed in the ChCl-MeOH system, having an 12.0 % dissolution. Furthermore, adding $AlCl_3$ ionized Cl^- increased the solvent's hydrogen bond acceptance and promoted hydrogen bonding's competitiveness to lignin while increasing the system's acidity to facilitate the breakage of lignin C-C and C-O bonds [27,28]. In the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ system with a molar ratio of 1:4:0.2, the lignin solubilization increased as the reaction conditions intensified (Fig. 1b). The lignin dissolution was 61.9 % at 100 °C–120 min, while the cellulose and hemicellulose retention were 74.7 % and 29.8 %, respectively. The lignin dissolution increased with temperature, and the cellulose retention reduced. However, the change in hemicellulose retention was insignificant, achieving a 95.0 % lignin dissolution at 120 °C–180 min, with 22.0 % and 45.0 % hemicellulose and cellulose retention, respectively.

The ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ molar ratio in this system was further investigated (Fig. 1c). As the relative content of MeOH in ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ increased, the lignin dissolution decreased gradually, while the cellulose and hemicellulose retention gradually increased until it stabilized. When the molar ratio of ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ was 1:12:0.2 (under the condition of 120 °C–180 min), its lignin dissolution was 93.1 %, while the cellulose and hemicellulose retention were 95.3 % and 47.8 %, respectively. We hypothesized that as the MeOH content of DES increased, the hydroxyl group could quickly form hydrogen bonds with cellulose components, thereby stabilizing the cellulose structure more efficiently.

The Kamlet-Taft solvent parameters can describe the DES nature, defined by three parameters: α, β, and π*. The α and β values are positively correlated with the ability of the solvent to give and receive hydrogen bonds, respectively, while π* is positively correlated with the solvent's polarity, i.e., the interactions between anions and cations in the solvent [29]. Among the binary DESs (Fig. 2a), ChCl-Buta and ChCl-OA had the largest values of α and β, corresponding to the largest lignin dissolution capacity, with the highest values of α (0.682) and β (0. 643) for the ChCl-OA system due to the strong hydrogen bonding of Cl-COOH. Hence, it quickly produced the off-domain phenomenon of H protons in this DES, increasing the acidity of the solvent with high hydrogen bond donor and acceptor, thereby greatly enhancing lignin's solubilization. While the α (0.205) and β (0.262) values of the ChCl-MeOH system were lower, the lower hydrogen bond supplying and accepting abilities were unfavorable for its binding with lignin. Then, when $AlCl_3{\cdot}6H_2O$ was added to the ChCl-MeOH system, the values of α and β increased compared to those of the original solvent system. This occurrence indicated that $AlCl_3{\cdot}6H_2O$ increased the acidity of the solvent and significantly improved the solvent's hydrogen supply and hydrogen bond-accepting abilities. Also, the lignin dissolution was considerably enhanced. In summary, the higher the α and β values of the DES system, the more favorable the lignin dissolution [19].

As shown in Fig. 2b, in the ternary system ChCl-MeOH-$AlCl_3{\cdot}6H_2O$, the π* value of the system gradually increased with the relative MeOH content, along with cellulose retention. The increase in the π* value may be attributed to the enhanced anionic and cationic interactions in the solvent system, capable of facilitating the association between anions and cations. Thus, it decreased their ability to form hydrogen bonds with the cellulose molecules, increasing cellulose retention. Whereas hemicellulose retention may be influenced by the solvent's acidity (Fig. 2c), the ChCl-OA system and the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ system had higher α values, indicating higher acidity, thereby favoring low hemicellulose retention.

3.2. Hydrogenolysis of Evodia lepta residue in the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ system

3.2.1. Effect of temperature and time on hydrogenolysis reactions

The main products of hydrogenolysis of *Evodia lepta* residue in the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ system vary with the reaction conditions used (Fig. 3a, c, and e). The phenolic product contents increased with the MeOH molar ratio. They showed an overall increasing trend before decreasing with increased temperature, in which the maximum content of phenolic products was 30.8 % at 110 °C–2 h for ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ with a molar ratio of 1:4:0.2, and reached the maximum values of 32.1 % and 37.7 % at 130 °C–4 h and 130 °C–2 h for molar ratios of 1:8:0.2 and 1:12:0.2, respectively. Increasing the MeOH molar ratio in ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ was unfavorable for accumulating ketone products. The total ketone product content increased with the reaction temperature for ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ molar ratios 1:4:0.2 and

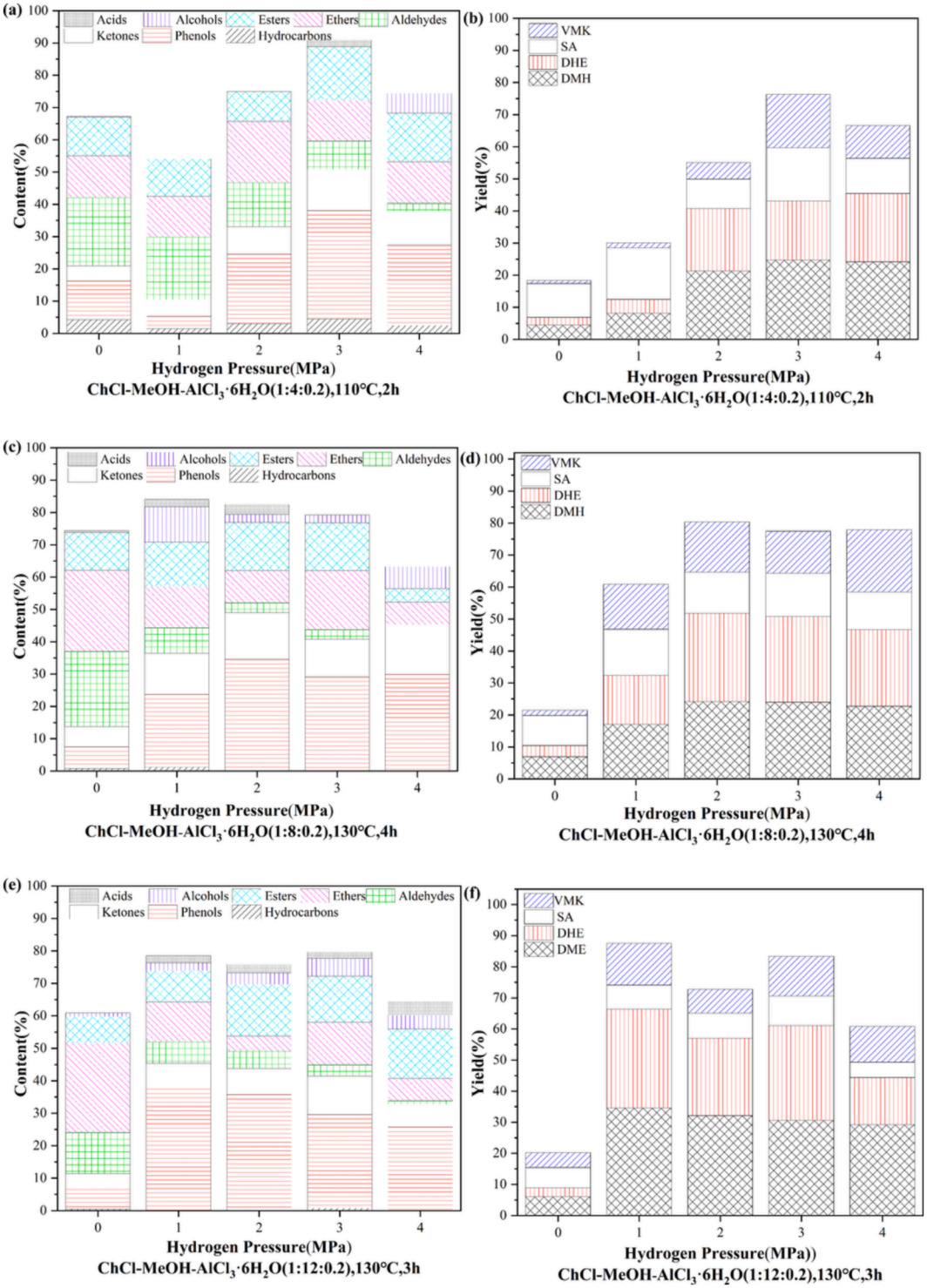


Fig. 4. Effect of hydrogen pressure on the distribution (a), (c), (e), yield (b), (d), and (f) depolymerization products in the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ system with different mole ratios.

1:8:0.2, reaching the maximum values of 32.8 % and 19.8 % at 150 °C–3 h and 150 °C–4 h, respectively. The MeOH- $AlCl_3{\cdot}6H_2O$ molar ratio of 1:12:0.2 reached the maximum value of 20.0 % at 90 °C–2 h.

For the total yield of phenolic and ketone small molecules (Fig. 3b, d, and f) for a DES molar ratio of 1:4:0.2 at 110 °C–2 h, the vanillyl methyl ketone (VMK), syringylacetone (SA), dihydroeugenol (DHE), and 4-propyl-2,6-dimethoxyphenol (DMH) yields were 16.6 %, 16.9 %, 17.8 %, and 23.6 %, respectively. These four phenol ketones were derived from lignin's guaiacyl (G) and silylene (S) monomers, which have the potential to be used in the synthesis of high-density jet fuel. Additionally, VMK is commonly used as a chemical synthesis intermediate, and DHE is widely used in pharmaceuticals due to its astringent and analgesic properties. The total yield reached a maximum value of 74.9 %, when the lignin dissolution was 91.9 %, and the hemicellulose and cellulose retention were 17.0 % and 86.2 % (Fig. S3a), respectively. The maximum total yields of phenolic and ketone small molecules at the DES molar ratios of 1:8:0.2 and 1:12:0.2 were 80.4 % (150 °C–3 h) and 85.7 % (150 °C–4 h), corresponding to 83.4 % (Fig. S3b) and 89.6 %

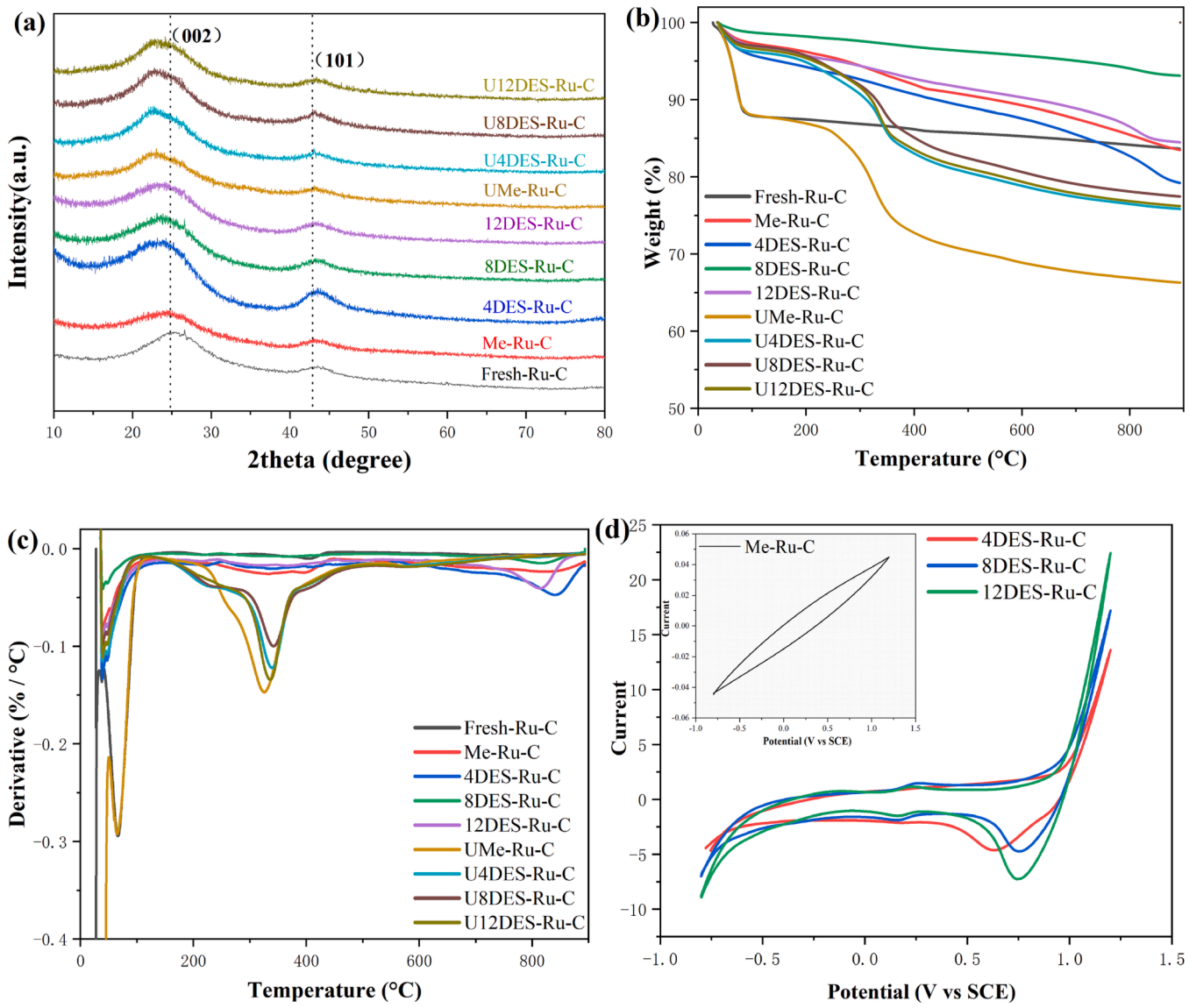


Fig. 5. Characterization of Ru-C samples: (a) XRD spectra, (b)-(c) TG-DTG spectra, (d) CV curves.

(Fig. S3c) lignin dissolution and 79.3 % (Fig. S3b) and 82.6 % (Fig. S3c) cellulose retention, respectively.

Alcohol products (such as hemicellulose-derived n-pentanol and tetrahydrofurfuryl alcohol) and methyl acetyl propionate from cellulose derivatives were detected in all hydrogenolysis products. Comparatively, the low (OH—Cl^-) binding due to ChCl component retention in DES under mild conditions significantly increased cellulose retention in the feedstock while ensuring high lignin dissolution [28,30]. It occurred where the total yield of phenolic and ketone small molecules under the DES molar ratio of 1:8:0.2 at 130 °C–4 h was 78.5 %, the lignin dissolution efficiency of the feedstock was 90.3 %, and the cellulose retention was as high as 96.2 % (Fig. S3b). The total yield of phenol ketone small molecules was 83.5 %, the raw lignin dissolution rate was 91.0 %, and the cellulose retention was as high as 94.1 % (Fig. S3c) under the DES molar ratio of 1:12:0.2 and 130 °C–3 h conditions. It achieved the purpose of preferential lignin conversion and high cellulose retention.

3.2.2. *Effect of hydrogen pressure on hydrogenolysis reactions*

Regardless of the molar ratio of ChCl-MeOH-$AlCl_3{\cdot}6H_2O$, the contents of phenolic of ChCl-MeOH-$AlCl_3{\cdot}6H_2O$, the contents of phenolic and ketone small molecule products increased at first before decreasing with increased gas pressure (Fig. 4a, c, and e). Here, the maximum phenolic small molecule content (33.6 %) appeared at the 3 MPa hydrogen pressure under the 1:4:0.2 M ratio. When the amount of MeOH in the DES system increased, the hydrogen donor of the system increased. The hydrogen pressure required for the maximum phenolic small molecule content appeared at a reduced time [31,32], 2 and 1 MPa for DES molar ratios of 1:8:0.2 and 1:12:0.2, respectively. However, this pattern was not evident in the ketone small molecule content, and the maximum ketone content of 12.7 %, 15.4 %, and 11.8 % under DES molar ratios of 1:4:0.2, 1:8:0.2, and 1:12:0.2 appeared at 3, 4, and 3 MPa, respectively. It suggested that the small molecules of ketone should be the primary products of lignin depolymerization, requiring more hydrogenolysis intensity to be generated [33,34]. This condition corresponded to the increased total ketone products with increased reaction temperature, as previously identified.

For the total yield of phenolic and ketone small molecules (Fig. 4b, d, and f), the VMK, SA, DHE, and DMH yields under DES molar ratio of 1:4:0.2, 3 MPa hydrogen pressure were 16.6 %, 16.6 %, 18.4 %, and 24.7 %, respectively. The maximum total yield was 76.3 %, lignin dissolution was 90.4 %, while hemicellulose and cellulose retentions were 25.2 % and 87. 4 % (Fig. S4a), respectively. The VMK, SA, DHE, and DMH yield under DES molar ratio of 1:8:0.2, 2 MPa hydrogen pressure were 15.7 %, 12.9 %, 27.6 %, and 24.2 %, respectively. Also,

the total product yield reached a maximum of 80.4 %, corresponding to a 90.3 % lignin dissolution, while 89.6 % (Fig. S4b) of the cellulose was retained. A further increase in the MeOH content of the DES would significantly increase the phenolic product yields. With a 1:12:0.2 DES molar ratio and at 1 MPa hydrogen pressure, the maximum total product yield was 87.5 %, in which the VMK, SA, DHE, and DMH yields were 13.4 %, 7.7 %, 31.8 %, and 34.6 %, respectively. This condition maintained cellulose retention up to 97.3 % while maintaining 91.3 % lignin dissolution (Fig. S4c).

Compared with temperature and time, hydrogen pressure, solid–liquid ratio (Fig. S1) and catalyst addition (Fig. S2) induced little effect on the hydrogenolysis. The effect of some of these factors on *Evodia lepta* residue fractionation were shown in Figs. S3–S6. Under the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$-DES molar ratio of 1:12:0.2, 130 °C, 3 h, 1 MPa, 1:20 g/mL, and 6:1 g/g (*Evodia lepta* residue: Ru-C), the total maximum yield of VMK, SA, DHE, and DMH reached 92.4 % (Fig. S2f), the corresponding total yields of 82.0 % (Fig. S2b) and 85.3 % (Fig. S2d) for DES molar ratios of 1:4:0.2 and 1:8:0.2, respectively. Compared with the reported 22.4 % yield of aromatic phenolic monomers using ethanol/1,4-dioxane/formic acid as solvent at 300 °C for 2 h [35], the *in-situ* hydrogenolysis of *Evodia lepta* residue using DES in the present study increased the aromatic monomer yield by 312.5 % under a more environmentally friendly and energy-saving condition.

3.3. Synergistic effect of DES with catalysts

3.3.1. XRD

The XRD spectra of Ru-C showed the degree of dispersion of Ru or Ru oxides on the carbon carriers. Two broad peaks at 25.4° and 44° (Fig. 5a) corresponded to the diffraction peaks on the (0 0 2) and (1 0 1) faces of the hexagonal crystalline carbon, respectively [36]. In contrast, no significant peak occurred for Ru (44° and 38° PDF-#06-0663) and RuO_2 (28°, 35°, and 54°, PDF-#43-1027), indicating that the active components of the ruthenium crystal size were small and highly dispersed in the catalyst [37]. Meanwhile, the crystallinity of Me-Ru-C (Fig. 5a) decreased at both sites compared with Fresh-Ru-C and DES-treated Ru-C. Also, the Ru-C crystallinity under various DES molar ratio systems was indifferent from that of Fresh-Ru-C, suggesting that the MeOH treatment would destroy the original crystal stability. On the other hand, the Ru-C crystal structure remained relatively intact after DES treatment, i.e., the crystal stability and crystallinity of Ru-C were maintained under the DES system.

3.3.2. TG

TG tested the thermal stability of the samples. The weight of 8DES-Ru-C and 12DES-Ru-C after the weight loss were higher than that of Fresh-Ru-C, suggesting that the DES with high methanol content better improved Ru-C's thermal stability. Moreover, the Ru-C weight was stabilized at about 76.0 % for DES system and 66.0 % for UMe-Ru-C after the reaction (Fig. 5b). Compared with the Ru-C treated by the DES system, the overall weight loss (DTG peak area) of UMe-Ru-C was the largest (Fig. 5c), inferring that the DES system more favorably maintained the thermal stability of Ru-C than MeOH. The main reason might be the effect of hydrogen bonding in the DES system, which lowers the melting and boiling points of solvents [38]. As a result, the compatibility with the catalyst compatibility, favorable for catalyst stabilization during the hydrogenolysis reaction, was improved.

3.3.3. Cyclic voltammetry

The redox potential of Ru-C treated with various solvents was investigated using cyclic voltammetry (CV). Ru-C was placed in methanol and different molar ratios of DES, resulting in 0, 4.4, 5.0, and 7.1 J/$mA{\cdot}cm^{-2}$ reduction potential for methanol, 1:4:0.2, 1:8:0.2, and 1:12:0.2 DES molar ratios, respectively (Fig. 5e). These results indicated that ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ had a better synergistic effect on Ru-C, effectively improving the reduction catalytic ability of Ru-C. With increased methanol content, the catalytic ability and depolymerization effect improved further, increasing the yield of aromatic phenolic ketone products. This result was consistent with those of hydrogenolysis. This phenomenon may be attributed to the electron transfer between DES and the catalyst, resulting in a change in the electronic structure of the catalyst [39]. The SEM further observed the microscopic morphology and surface structure of the catalysts. Compared to Me-Ru-C, the pore structure of 12DES-Ru-C was significantly increased, resulting in a more homogeneous dispersion of Ru particles (Figs. S7e and 6f), the relevant results can be found in the Supporting Information S3.

Table 1
Specific BET surface area (S_{BET}), average pore size, pore volume, and micropore volume of the catalysts.

Catalyst	S_{BET}[a] (m^2/g)	V[b] (cm^3/g)	Dv[b] (nm)	V_{micro}[c] (cm^3/g)
Fresh-RU-C	1173	0.696	2.373	0.548
4DES-Ru-C	1092	0.550	2.015	0.434
8DES-Ru-C	1113	0.563	2.023	0.442
12DES-Ru-C	1122	0.601	2.143	0.448
Me-Ru-C	1086	0.580	2.136	0.433
U4DES-Ru-C	320.1	0.307	3.839	0.122
U8DES-Ru-C	317.7	0.154	1.941	0.104
U12DES-Ru-C	435.6	0.210	1.924	0.135

Wherein 4DES-Ru-C, 8DES-Ru-C, and 12DES-Ru-C correspond to Ru-C after immersion treatment under the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ (1:4:0.2, 1:8:0.2, 1:12:0.2) system, respectively. U4DES-Ru-C, U8DES-Ru-C, and U12DES-Ru-C correspond to Ru-C after hydrogenolysis in the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ (1:4:0.2, 1:8:0.2, and 1:12:0.2) system, respectively. Fresh-Ru-C was the untreated Ru-C.

[a] Total specific surface area was calculated by the BET equation.

[b] The pore volume and average pore diameter of mesopores were calculated by the BJH method.

[c] The micropore volume was calculated by HK.

Wherein 4DES-Ru-C, 8DES-Ru-C, and 12DES-Ru-C correspond to Ru-C after immersion treatment under the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ (1:4:0.2, 1:8:0.2, 1:12:0.2) system, respectively. U4DES-Ru-C, U8DES-Ru-C, and U12DES-Ru-C correspond to Ru-C after hydrogenolysis in the ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ (1:4:0.2, 1:8:0.2, and 1:12:0.2) system, respectively. Me-Ru-C was the Ru-C soaked in methanol solvent and Fresh-Ru-C was the untreated Ru-C.

3.3.4. BET

The results in Fig. S7a and Table 1 showed that the unreacted Ru-C series were predominantly microporous structures with pore volumes ranging from 0.433 to 0.548 cm^3/g. They were accompanied by many mesoporous structures with a 2 nm average pore size. The specific surface area, pore volume, average pore size, and microporous pore volume of the MeOH and DES-treated Ru-C series were smaller than those of Fresh-Ru-C. This difference may be ascribed to the dissolution collapse of some of the Ru-C pore walls in the presence of solvents [40]. Moreover, the DES-treated series exhibited a larger specific surface area and pore size than the MeOH-treated Ru-C. Along with the increase in DES molar ratio, the specific surface area, average pore size, and pore volume of Ru-C increased. Similarly, the specific surface area of U4DES-Ru-C, U8DES-Ru-C, and U12DES-Ru-C also increased after the reaction. It has been suggested that it is due to the hydrogen bonding effect of DES that improves Ru dispersion. High specific surface area provides more catalytically active sites and larger pore size (based on spatial constraints), preventing the recondensation of highly active intermediates [27]. An increased pore volume would provide a more suitable reaction site for larger reactive substrates [27], consistent with the increase in reaction effectiveness accompanying increased DES molar ratio in the hydrogenolysis reaction. Compared with the pre-reaction period, the specific surface area and pore volume of the various solvent-treated Ru-C were smaller after the reaction. The generation of deposited carbon rationalized this occurrence during the organic reaction process [41].

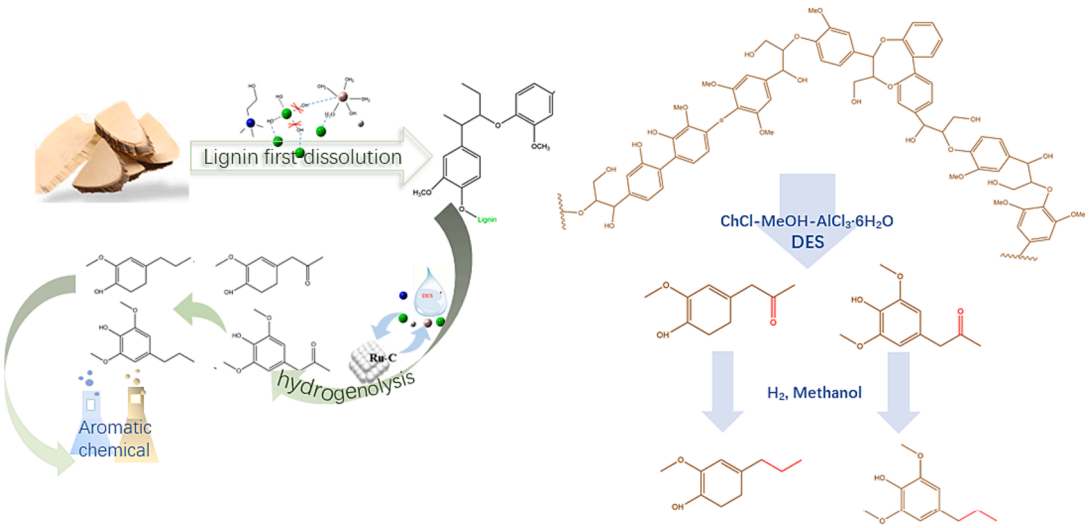


Fig. 6. The depolymerization mechanism of lignin in ChCl-MeOH-$AlCl_3{\cdot}6H_2O$.

3.3.5. *The depolymerization mechanism*

This paper analyzed and investigated the mechanism of catalytic hydrogenolysis of *Evodia lepta* residue by the DES system. The lignin in the *Evodia lepta* residue was dissolved under the DES system (Fig. 6) before it was catalytically depolymerized under the joint action of ChCl-MeOH and $AlCl_3{\cdot}6H_2O$. Here, the bonds in the depolymerized lignin structure (especially the β-O-4 ether bond) were broken by Ru-C catalysts, generating lignin aromatic compounds by selective reduction under the sufficient hydrogen atmosphere [42]. The ketones vanillyl methyl ketone and butylacetone were selectively reduced under adequate hydrogen atmosphere to form lignin aromatic ketones (such as (E)-2-methoxy-4-(1-propenylphenol), (E)-4-(3-hydroxyprop-1-enyl)-2-methoxyphenol, and 3-(4-hydroxy-3-methoxyphenyl)-1-propanol, etc.) along with catalyzed hydrogenolysis.

The intermediates in the transition form (e.g., butylacetone) were gradually reduced to form (E)-4-(3-hydroxyprop-1-enyl)-2,6-dimethoxyphenol, 2,6-dimethoxy-4-[(1E)-1-propen-1-yl] phenol, and 4-propyl-2,6-dimethoxyphenol. Also, these products were further reduced or dehydrated into lignin aromatic compounds in the hydrogen atmosphere. As a result, the content of ketone products and the yields of VMK and SA gradually decreased. In contrast, the content of phenolic products and the yields of DMH and DHE compounds gradually increased.

On the other hand, due to the increase in methanol as a hydrogen donor, the hydrogen pressure of the system increased, and its hydrogenolysis capacity further increased [43]. Thus, the reduction reaction ensued more thoroughly, thus accelerating the depolymerization efficiency and the conversion between aromatic keto-phenolic compounds [44]. Consequently, these aromatic compounds can be used as feedstock for synthesizing high density jet fuel while changing DES molar ratio regulates the product type and distribution.

4. Conclusion

The ChCl-MeOH-$AlCl_3{\cdot}6H_2O$ DES system developed in this study showed superior performance for *in-situ* hydrogenolysis of lignin from *Evodia lepta* residue with the high yield of vanillyl methyl ketone, syringylacetone, dihydroeugenol, and 4-propyl-2,6-dimethoxyphenol. And the maximum total aromatic phenol ketone yield was 92.4 %, which was 312.5 % higher than that of the reported methods. The coupling of DES with Ru-C formed a more efficient hydrogenolysis system, which improved the thermal stability, specific surface area and Ru-C reduction potential while maintaining its crystallinity. To our knowledge, this is the first report of a high yield of aromatic phenol ketones from *Evodia lepta* residue.

CRediT authorship contribution statement

Shutian Chen: Writing – original draft, Investigation, Formal analysis, Data curation. **Yating Jian:** Methodology, Investigation, Data curation. **Yaowei Liu:** Investigation. **Zhanying Zhang:** Writing – review & editing, Visualization, Validation. **Xinlong Li:** Investigation. **Qiang Yu:** Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2024.131348.

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